■ Supplementary document

Sound, vibrations that cross the air and that one day made it possible to heal hearts. That's the story we're going to tell. Close your eyes and hold out your ear. The wind, a voice, a slamming of the door. Everything you hear comes from something moving. A sound is always born of a vibration. A trembling guitar string, a loudspeaker diaphragm that moves forward and backward, your vocal cords wiggling. Nothing that vibrates, nothing that rings. Experience it. Put two fingers on your throat and talk. Can you feel the tremor? It's the sound before you've even left your mouth. This vibration does not stay put. It is transmitted from molecule to molecule, as a wave crosses the sea. It's a wave. The air is not empty. It contains billions of billions of molecules. The sound pushes them, it tightens and then moves away and it is this back and forth that carries the information. Be careful, this is essential. A sound needs a material medium to propagate. No air, no water, no metal and the wave has nothing left to travel. In space, no one will hear you scream. It's not a matter of distance. There is simply no air to carry the vibrations. In the air, the sound flies at 340 meters per second. It's fast. In a second, he crosses a football field and a half. And yet, light goes 880,000 times faster. That's why you see the lightning before you hear the thunder. The sound, on the other hand, takes its time. The sound also does not go at the same speed depending on the medium. 1500 meters per second in water, 5000 in steel. The tighter the molecules, the faster the wave travels. Two quantities are enough to describe a sound. Its frequency, measured in hertz, which gives the height. And its amplitude, which gives the volume, that is to say the strength of the vibration. A high frequency gives a high-pitched sound, like the whistle. A low frequency gives a low sound, like the bass drum. The higher the sound, the faster the vibration. The amplitude is the extent of the back and forth. The wider the vibration, the louder the sound. This is what we measure in decibels, and it is also what damages the ears. There remains a third thing, the most mysterious, the stamp. Play the same note on piano and guitar, you immediately recognize both instruments. It's the stamp. In your ear, these vibrations become a signal. The eardrum begins to vibrate, three tiny bones amplify the movement and the message goes to the brain. But your ear is limited. It only hears between about 20 and 20,000 hertz. Below or above, the sound still exists. You just can't hear it. Below 20 hertz, it's the infrassons. Elephants use them to communicate from miles away, vibrating the ground beneath their paws. Above 20,000 hertz, it's ultrasound. Bats use it to find their way around in the dark. They scream and listen to the echo that comes back. An echo is simply a sound that bounces off an obstacle. Boat sonars and hospital ultrasounds use exactly this principle. This is where Mickaël Tanter, a French physicist, comes in. His playground, those ultrassons that nobody hears. His idea, send ultrasounds into the body, then listen to what he sends back. These are not X-rays. Here, the image is made with sound. A conventional ultrasound produces about fifty images per second. The technique they develop produces thousands. We are talking about ultra-fast ultrasound. The consequence is enormous. We can film the blood circulating in the brain in real time, without opening, without irradiating and without injecting contrast media. This technology gave birth to Iconeus, a startup whose machines now equip hospitals to help diagnose brain diseases. But Mickaël Tanter then decides to tackle even more difficult issues. The heart. An organ that beats non-stop, so very difficult to follow and treat. The problem he wants to solve has a name, aortic shrinkage. With age, a heart valve hardens and lets less and less blood through. Today, the solution is a heavy operation. The chest is opened to replace the valve. It's effective, but risky and time-consuming to recover. His idea is radical. Replace the scalpel with waves. Focused ultrasound, focused on the valve, which softens the diseased tissue without cutting anything. The principle already exists elsewhere. Kidney stones are broken with shock waves, without ever operating. It remains to do the same on a moving organ. But the heart is much more complex than a kidney. For 15 years, the idea seems too fragile and few people really believe it. Yet, a clinical trial conducted on 40 patients shows that the technique is safe and that it improves the condition of patients. In 2023, these results were published in The Lancet, one of the most widely read medical journals in the world. International recognition. The start-up CardiaWave is created and obtains the CE mark. His device can now be used for healing in Europe. The benefit is simple, treat the heart without opening it. Less risk to the patient, shorter hospital stay and much faster recovery. There are still steps and they are serious. Multiply trials, treat more patients and mass-produce machines. What you need to remember is in three words. A vibration, a medium, a frequency. Initially, it's just simple physics, that of college. And on arrival, a tool that heals. Science is not just formulas. This is what we do with it and it is often very concrete. So the next time you listen to music, remember. You hear vibrations that have traveled to you and will one day repair hearts. If you liked this video, leave a like and subscribe so you don't miss anything next.